

Assignment 3

GPT-4.1, the Worst
Mathematician Ever

CART 498B

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The Worst Mathematician Ever

The goal of my AI bot was to test how well a GPT-4.1 model could act as a math calculator and identify mathematical operations that would make it fail. For calculations such as $8*8$, 5^5 , and 8^8 , the bot performed well and the results that I received matched Python's results. However, as the calculations became bigger, containing larger numbers, it clearly became more complex for the bot and it started making mistakes.

GPT models are primarily language models and perform better with text prompts, not mathematical models. Keeping in mind that these models also have a text memory, they perform better with text-based tasks, rather than exact math calculations. Although they can often solve common, and rather simple calculations accurately, they are not able to deal with very large numbers.

A second issue appeared with exponentiation. In my code, Python converts $\wedge$ into $**$ before evaluating the expression, but the AI only sees the user's original input. When the model received expressions such as 125^{125} , it returned "ERROR" instead of a number. The model misinterpreted what I was asking or considered the calculation too difficult. Since $\wedge$ is not universally used for exponentiation in programming languages, this may have confused the model. This had to be corrected and the following had to be implemented: `python_expression = user_input.replace('^', '**')` for the model to be able to understand exponentiation.

The most interesting failures occurred when expressions combined large exponents with additional operations. For example, $25^{25+25*25}$ and 8^8+8*11 produced completely incorrect answers. In these cases, the model likely lost track of the order of operations.

Overall, the experiment demonstrated that GPT-4.1 can function as a surprisingly capable calculator for many math problems, but it is not perfect as one may think. Its failures occurred primarily with very large exponentiation problems and more complex expressions, confirming that language models should not be trusted as exact calculators without verifying the results

<https://github.com/NicolasH21/CART498-GenAI>